Table 1. Properties of Some Ideal Gases

Gas	Symbol	m_{wt} g/mol	R J/kg-K	c_{po} J/kg-K	c_{vo} J/kg-K	γ
Air: $0.21O_2 + 0.79N_2$		28.97	287.00	1005	718.0	1.4
Oxygen	O_2	32.01	259.7	921.6	661.8	1.4
Nitrogen	N_2	28.013	296.80	1041.6	744.8	1.4
Carbon dioxide	CO_2	44.01	188.92	841.8	652.9	1.289
Helium	He	4.003	2077.03	5192.6	3115.6	1.667
Hydrogen	H_2	2.016	4124.18	14209.1	10084.9	1.409
Propane	C_3H_8	44.097	188.55	1679.4	1490.9	1.126
Carbon Monoxide	CO	28.0	296.9	1035.7	738.8	1.4
Water vapor	H_2O	18.015	461.52	1872.3	1410.8	1.327

Table 2. Liquid Fuel Property Data

Fuel	Density Kg/m^3	LHV MJ/kg	LHV MJ/litre	CO_2-kg/MJ
Aviation Gasoline (avgas)	708.2	43.7	33.0	0.068
Aviation Turbine Fuel (Jet A1)	793.0	43.1	36.8	0.0697

Table 3. Planetary Data for the Earth

Quantity	Value
Gravity	$9.81 m/s^2$
Escape Velocity	11.2km/s
Average Surface Radius	6378 km
Mean Molecular Weight	28.97 g/mol
Mean Gas Constant	287J/kg-K

Table 4. Typical Composition of the Earth's Atmosphere at Sea-level

Gas	Symbol	% (Mass)
Nitrogen	N_2	78.09
Oxygen	O_2	20.95
Argon	Ar	0.93
Carbon dioxide	CO_2	0.02 (to 0.04)
Neon	Ne	0.0018
Helium	He	0.0005
Krypton	Kr	0.0001
Hydrogen	H_2	0.00005
Xenon	Xe	0.000008
Ozone	O_3	0.000002
Radon	Rn	<0.000001

Table 5. International Standard Atmosphere (1976)

Altitude Km	Temperature K	Pressure Pa	Density Kg/m^3	Speed of Sound m/s
0	288.15	101325	1.225	340.2941
.5	284.9	95460.8393	1.1673	338.3696
1	281.65	89874.5705	1.1116	336.434
1.5	278.4	84556.0048	1.0581	334.4873
2	275.15	79495.2155	1.0065	332.5292
2.5	271.9	74682.5337	.9569	330.5595
3	268.65	70108.5447	.9091	328.578
3.5	265.4	65764.0844	.8632	326.5845
4	262.15	61640.2353	.8191	324.5787
4.5	258.9	57728.323	.7768	322.5604
5	255.65	54019.9121	.7361	320.5295
5.5	252.4	50506.803	.6971	318.4856
6	249.15	47181.0276	.6597	316.4284
6.5	245.9	44034.8461	.6238	314.3579
7	242.65	41060.7432	.5895	312.2736
7.5	239.4	38251.4241	.5566	310.1752
8	236.15	35599.8114	.5252	308.0626
8.5	232.9	33099.0409	.4951	305.9355
9	229.65	30742.4584	.4663	303.7934
9.5	226.4	28523.6158	.4389	301.6361
10	223.15	26436.2676	.4127	299.4632
10.5	219.9	24474.3673	.3877	297.2745
11	216.65	22632.064	.3639	295.0696
11.5	216.65	20916.189	.3363	295.0696
12	216.65	19330.405	.3108	295.0696
12.5	216.65	17864.849	.2873	295.0696
13	216.65	16510.4058	.2655	295.0696
13.5	216.65	15258.6511	.2454	295.0696
14	216.65	14101.7996	.2268	295.0696
14.5	216.65	13032.6561	.2096	295.0696
15	216.65	12044.5709	.1937	295.0696
15.5	216.65	11131.3984	.179	295.0696
16	216.65	10287.4591	.1654	295.0696
16.5	216.65	9507.5041	.1529	295.0696
17	216.65	8786.6821	.1413	295.0696
17.5	216.65	8120.5101	.1306	295.0696
18	216.65	7504.8447	.1207	295.0696
18.5	216.65	6935.8566	.1115	295.0696
19	216.65	6410.0069	.1031	295.0696
19.5	216.65	5924.0252	0.0953	295.0696
20	216.65	5474.8887	0.088	295.0696
20.5	217.15	5060.2636	0.0812	295.4099
21	217.65	4677.8862	0.0749	295.7498
21.5	218.15	4325.1827	0.0691	296.0893
22	218.65	3999.7902	0.0637	296.4284
22.5	219.15	3699.5385	0.0588	296.7671

Altitude Km	Temperature K	Pressure Pa	Density Kg/m^3	Speed of Sound m/s
23	219.65	3422.4343	0.0543	297.1055
23.5	220.15	3166.6465	0.0501	297.4434
24	220.65	2930.4922	0.0463	297.781
24.5	221.15	2712.425	0.0427	298.1182
25	221.65	2511.0234	0.0395	298.455
25.5	222.15	2324.9802	0.0365	298.7915
26	222.65	2153.0938	0.0337	299.1275
26.5	223.15	1994.2584	0.0311	299.4632
27	223.65	1847.4574	0.0288	299.7985
27.5	224.15	1711.7549	0.0266	300.1335
28	224.65	1586.2898	0.0246	300.468
28.5	225.15	1470.2697	0.0227	300.8022
29	225.65	1362.9648	0.021	301.136
29.5	226.15	1263.7033	0.0195	301.4695
30	226.65	1171.8665	0.018	301.8026

Table 6. Escape Velocities of Various Planets and the Moon

Planet	g_0 m/s^2	r_o km	U_{esc} km/s
Mercury	3.53	2487	4.19
Venus	8.87	6200	10.4
Earth	9.91	6378	11.2
Moon	1.67	1740	2.41
Mars	3.69	3412	5.03
Jupiter	25.9	71434	60.8
Saturn	11.1	60400	36.6
Uranus	10.2	23535	21.9
Neptune	13.8	2233	24.8

Table 7. Planetary Data for Mars

Quantity	Value
Gravity	3.69 m/s^2
Escape Velocity	5.03 km/s
Surface Pressure	610 Pa
Surface Density	0.02 kg/m^3
Scale Height	11.1km
Average Temperature	210 K
Mean Molecular Weight	43.34g /mole
Mean Gas Constant	0.1918 kJ/kgK
Average Radius	3412km

Table 8. Typical composition of Mars' Atmosphere at Surface Level

Gas	Value
Major	*%*
CO_2	95.32
N_2	2.7
Ar	1.6
O_2	0.13
CO	0.08
Minor	*PPM*
H_2O	210
NO	100
Ne	2.5
HDO	0.85
Kr	0.3
Xe	0.08

Table 9. Planetary Data for Venus

Quantity	Value
Gravity	8.87 m/s^2
Escape Velocity	10.36 km/s
Surface Pressure	9200 kPa
Surface Density	65 kg/m^3
Scale Height	15.9 km
Average Temperature	737 K
Mean Molecular Weight	43.45 g/mole
Mean Gas Constant	0.1914 kJ/kgK
Average Radius	6200km

Table 10. Typical composition of Venus' Atmosphere at Surface Level

Gas	Value
Major	*%*
CO_2	96.5
N_2	3.5
Minor	*PPM*
SO_2	150
Ar	70
H_2O	20
CO	17
Ne	7

Table 11. Fuel properties

Table B.1 Selected properties of hydrocarbon fuels: enthalpy of formation,[a] Gibbs function of formation,[a] entropy,[a] and higher and lower heating values all at 298.15 K and 1 atm; boiling points[b] and latent heat of vaporization[c] at 1 atm; constant-pressure adiabatic flame temperature at 1 atm;[d] liquid density[c]

Formula	Fuel	MW (kg/kmol)	$\bar{h}_f^\circ$ (kJ/kmol)	$\bar{g}_f^\circ$ (kJ/kmol)	$\bar{s}^\circ$ (kJ/kmol-K)	HHV† (kJ/kg)	LHV† (kJ/kg)	Boiling pt. (°C)	h_{fg} (kJ/kg)	$T_{ad}^\ddagger$ (K)	ρ_{liq}^* (kg/m³)
CH_4	Methane	16.043	−74,831	−50,794	186.188	55,528	50,016	−164	509	2,226	300
C_2H_2	Acetylene	26.038	226,748	209,200	200.819	49,923	48,225	−84	—	2,539	—
C_2H_4	Ethene	28.054	52,283	68,124	219.827	50,313	47,161	−103.7	—	2,369	—
C_2H_6	Ethane	30.069	−84,667	−32,886	229.492	51,901	47,489	−88.6	488	2,259	370
C_3H_6	Propene	42.080	20,414	62,718	266.939	48,936	45,784	−47.4	437	2,334	514
C_3H_8	Propane	44.096	−103,847	−23,489	269.910	50,368	46,357	−42.1	425	2,267	500
C_4H_8	1-Butene	56.107	1,172	72,036	307.440	48,471	45,319	−63	391	2,322	595
C_4H_{10}	*n*-Butane	58.123	−124,733	−15,707	310.034	49,546	45,742	−0.5	386	2,270	579
C_5H_{10}	1-Pentene	70.134	−20,920	78,605	347.607	48,152	45,000	30	358	2,314	641
C_5H_{12}	*n*-Pentane	72.150	−146,440	−8,201	348.402	49,032	45,355	36.1	358	2,272	626
C_6H_6	Benzene	78.113	82,927	129,658	269.199	42,277	40,579	80.1	393	2,342	879
C_6H_{12}	1-Hexene	84.161	−41,673	87,027	385.974	47,955	44,803	63.4	335	2,308	673
C_6H_{14}	*n*-Hexane	86.177	−167,193	209	386.811	48,696	45,105	69	335	2,273	659
C_7H_{14}	1-Heptene	98.188	−62,132	95,563	424.383	47,817	44,665	93.6	—	2,305	—
C_7H_{16}	*n*-Heptane	100.203	−187,820	8,745	425.262	48,456	44,926	98.4	316	2,274	684
C_8H_{16}	1-Octene	112.214	−82,927	104,140	462.792	47,712	44,560	121.3	—	2,302	—
C_8H_{18}	*n*-Octane	114.230	−208,447	17,322	463.671	48,275	44,791	125.7	300	2,275	703
C_9H_{18}	1-Nonene	126.241	−103,512	112,717	501.243	47,631	44,478	—	—	2,300	—
C_9H_{20}	*n*-Nonane	128.257	−229,032	25,857	502.080	48,134	44,686	150.8	295	2,276	718
$C_{10}H_{20}$	1-Decene	140.268	−124,139	121,294	539.652	47,565	44,413	170.6	—	2,298	—
$C_{10}H_{22}$	*n*-Decane	142.284	−249,659	34,434	540.531	48,020	44,602	174.1	277	2,277	730
$C_{11}H_{22}$	1-Undecene	154.295	−144,766	129,830	578.061	47,512	44,360	—	—	2,296	—
$C_{11}H_{24}$	*n*-Undecane	156.311	−270,286	43,012	578.940	47,926	44,532	195.9	265	2,277	740

Table 12. Formation Enthalpies

Standard enthalpies of formation

Species	State†	$\Delta\bar{h}_f^\circ$, MJ/kmol
O_2	Gas	0
N_2	Gas	0
H_2	Gas	0
C	Gas	0
CO_2	Gas	−393.52
H_2O	Gas	−241.83
H_2O	Liquid	−285.84
CO	Gas	−110.54
CH_4	Gas	−74.87
C_3H_8	Gas	−103.85
CH_3OH	Gas	−201.17
CH_3OH	Liquid	−238.58
C_8H_{18}	Gas	−208.45
C_8H_{18}	Liquid	−249.35

† At 298.15 K (25°C) and 1 atm.

Table 13. Liquid Fuel/Oxidiser Propellant Data

Fuel	Fuel density [kg/m^3]	Oxidiser	Oxidiser density [kg/m^3]	Products Molecular Weight [g/mol]	Products Temperature [K]
H_2	70.85	O_2	1141	10.08	2980
H_2	70.85	F_2	1505	13.64	4117
H_2	70.85	F_2O	1521	11.36	3589
H_2	70.85	ClF_3	1770	16.78	3390
N_2H_4 / Hydrazine	1032	O_2	1141	20.29	3410
N_2H_4 / Hydrazine	1032	F_2	1505	21.33	3917
RP1 / Kerosene	810	O_2	1141	25.29	3623
RP1 / Kerosene	810	F_2	1505	25.26	3917

Conversion and Reference Data

1 knot = 1.85 kph Reference Temperature 298K R_u=8.314 kJ/kmol.K

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State condition numbering used in equations is arbitrary.

$$S = bc \qquad A = \frac{b}{c} \qquad \mathrm{V_{cal}} = \mathrm{V_E} = \mathrm{V_0}\sqrt{\frac{\rho}{\rho_s}} = \mathrm{V_0}\sqrt{\sigma} \qquad \eta_{th} = 1 - \left(\frac{1}{r_v}\right)^{\gamma-1}$$

$$L = \frac{1}{2}\rho U^2 c_L S \qquad D = \frac{1}{2}\rho U^2 c_D S \qquad c_L = a\alpha \qquad \eta_p = \frac{1}{1 + w/V}$$

$$\dot{W}_I = (IMEP)\dot{V}_s \qquad \dot{V}_s = A_P L_S \frac{N}{2} n \qquad \eta = \frac{\dot{W}_n}{\dot{Q}_H} \qquad \dot{m} = \rho V A$$

$$TSFC = \frac{\dot{m}_f}{F_N} \qquad BSFC = \frac{\dot{m}_f}{\dot{W}_b} \qquad F_s = \frac{F_N}{\dot{m}_a} \qquad f = \frac{\dot{m}_f}{\dot{m}_a}$$

$$\dot{W}_b = F_T(w+u) \qquad w = \frac{1}{2}\left[\left(u^2 + \frac{2F_T}{\rho A_p}\right)^{1/2} - u\right] \qquad \dot{W}_{b,o} = F_{To}{}^{3/2}/\sqrt{2\rho A_p}$$

$$p = \rho RT \qquad pv = RT \qquad \gamma = C_p/C_v \qquad R = C_p - C_v$$

$$(pv^\gamma)_1 = (pv^\gamma)_2 \qquad (Tv^{\gamma-1})_1 = (Tv^{\gamma-1})_2$$

$$\sum Q + \sum W = 0 \qquad Q + W = \Delta U \qquad LHV = \left[(h_f^o)_R - (h_f^o)_P\right]/M_{wf}$$

$$\left(\frac{p}{\rho}+u+\frac{C^2}{2}\right)_1=\left(\frac{p}{\rho}+u+\frac{C^2}{2}\right)_2+Q+W$$

$$\left(\frac{p}{\rho}+u\right)=h \qquad (h_2-h_1)=C_p(T_2-T_1) \qquad h=C_p(T-T_R)$$

$$\frac{p_0}{p}=\left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}} \qquad \frac{p_{02}}{p_{01}}=\left(\frac{T_{02'}}{T_{01}}\right)^{\frac{\gamma}{\gamma-1}} \qquad \eta_i=\frac{T_{01'}-T_a}{T_{01}-T_a}$$

$$T_{0a}=T_a+\frac{C_a^2}{2c_p} \qquad \frac{p_{01}}{p_a}=\left[1+\eta_i\frac{C_a^2}{2c_pT_a}\right]^{\frac{\gamma_a}{\gamma_a-1}} \qquad TP=\dot{W}_s\eta_p+C_aF_N \qquad \eta_c=\frac{T_{02'}-T_{01}}{T_{02}-T_{01}}$$

$$\eta_t=\frac{T_{03}-T_{04}}{T_{03}-T_{04'}} \qquad \eta_j=\frac{T_{04}-T_5}{T_{04}-T_{5'}} \qquad \frac{p_{04}}{p_c}=\left[1-\frac{1}{\eta_j}\left(\frac{\gamma-1}{\gamma+1}\right)\right]^{-\left(\frac{\gamma}{\gamma-1}\right)} \qquad \frac{T_c}{T_{04}}=\left(\frac{2}{\gamma+1}\right)$$

$$M_{n,a}=M_a\sin\beta \qquad M_1=M_{n,a}/\sin(\beta-\theta)$$

$$a=\sqrt{R\gamma T} \qquad M=\frac{C}{a} \qquad B=\frac{\dot{m}_c}{\dot{m}_h} \qquad FPR=p_{02}/p_{01} \qquad OPR=p_{03}/p_{01}$$

$$C_j=\left[\left(\frac{2\gamma}{\gamma-1}\right)RT_c\left(1-\left(\frac{p_j}{p_c}\right)^{(\gamma-1)/\gamma}\right)\right]^{1/2} \qquad \frac{A}{A^*}=\frac{1}{M}\left(\frac{2+(\gamma-1)M^2}{\gamma+1}\right)^{(\gamma+1)/(2\gamma-2)}$$

$$\Delta C=C_e\ln(M_R)-gt_b\sin\theta \qquad z_b=C_ot_o+C_et_b\left(\frac{1}{M_R-1}\ln\left(\frac{1}{M_R}\right)+1\right)-\frac{gt_b^2\sin\theta}{2}$$

$$z_c=z_b+\frac{r_oC_b^2}{2g_or_o-C_b^2} \qquad M_R=\frac{M_o}{M_f} \qquad C_{esc}=r_o\sqrt{\left(\frac{2g_o}{r_o+z}\right)} \qquad C_{sat}=r_o\sqrt{\left(\frac{g_o}{r_o+z}\right)}$$

Table 14. Normal Shock Tables

TABLE E.4 Normal Shock Flow of an Ideal Gas ($k = 1.4$)

Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.00	1.00000	1.00000	1.00000	1.00000	1.00000	1.8929	1.49	0.70466	2.4234	1.3136	1.8449	0.93292	3.3756
1.01	0.99013	1.02345	1.00665	1.01669	0.99999	1.9152	1.50	0.70109	2.4583	1.3202	1.8621	0.92978	3.4133
1.02	0.98052	1.04713	1.01325	1.03344	0.99998	1.9379	1.51	0.69758	2.4934	1.3269	1.8792	0.92658	3.4512
1.03	0.97115	1.07105	1.01981	1.05024	0.99997	1.9610	1.52	0.69413	2.5288	1.3336	1.8962	0.92331	3.4894
1.04	0.96202	1.09520	1.02634	1.06709	0.99994	1.9845	1.53	0.69073	2.5644	1.3403	1.9133	0.91999	3.5279
1.05	0.95312	1.1196	1.03284	1.08398	0.99987	2.0083	1.54	0.68739	2.6003	1.3470	1.9303	0.91662	3.5667
1.06	0.94444	1.1442	1.03931	1.10092	0.99976	2.0325	1.55	0.68410	2.6363	1.3538	1.9473	0.91319	3.6058
1.07	0.93598	1.1690	1.04575	1.11790	0.99962	2.0570	1.56	0.68086	2.6725	1.3606	1.9643	0.90970	3.6451
1.08	0.92772	1.1941	1.05217	1.13492	0.99944	2.0819	1.57	0.67768	2.7090	1.3674	1.9812	0.90615	3.6847
1.09	0.91965	1.2194	1.05856	1.15199	0.99921	2.1072	1.58	0.67455	2.7458	1.3742	1.9981	0.90255	3.7245
1.10	0.91177	1.2450	1.06494	1.1691	0.99892	2.1328	1.59	0.67147	2.7828	1.3811	2.0149	0.89889	3.7645
1.11	0.90408	1.2708	1.07130	1.1862	0.99858	2.1588	1.60	0.66844	2.8201	1.3880	2.0317	0.89520	3.8049
1.12	0.89656	1.2968	1.07764	1.2034	0.99820	2.1851	1.61	0.66545	2.8575	1.3949	2.0485	0.89144	3.8456
1.13	0.88922	1.3230	1.08396	1.2206	0.99776	2.2118	1.62	0.66251	2.8951	1.4018	2.0652	0.88764	3.8866
1.14	0.88204	1.3495	1.09027	1.2378	0.99726	2.2388	1.63	0.65962	2.9330	1.4088	2.0820	0.88380	3.9278
1.15	0.87502	1.3762	1.09657	1.2550	0.99669	2.2661	1.64	0.65677	2.9712	1.4152	2.0986	0.87992	3.9693
1.16	0.86816	1.4032	1.10287	1.2723	0.99605	2.2937	1.65	0.65396	3.0096	1.4228	2.1152	0.87598	4.0111
1.17	0.86145	1.4304	1.10916	1.2896	0.99534	2.3217	1.66	0.65119	3.0482	1.4298	2.1318	0.87201	4.0531
1.18	0.85488	1.4578	1.11544	1.3069	0.99455	2.3499	1.67	0.64847	3.0870	1.4369	2.1484	0.86800	4.0954
1.19	0.84846	1.4854	1.12172	1.3243	0.99371	2.3786	1.68	0.64579	3.1261	1.4440	2.1649	0.86396	4.1379
1.20	0.84217	1.5133	1.1280	1.3416	0.99280	2.4075	1.69	0.64315	3.1654	1.4512	2.1813	0.85987	4.1807
1.21	0.83601	1.5414	1.1343	1.3590	0.99180	2.4367	1.70	0.64055	3.2050	1.4583	2.1977	0.85573	4.2238
1.22	0.82998	1.5698	1.1405	1.3764	0.99073	2.4662	1.71	0.63798	3.2448	1.4655	2.2141	0.85155	4.2672
1.23	0.82408	1.5984	1.1468	1.3938	0.98975	2.4961	1.72	0.63545	3.2848	1.4727	2.2304	0.84735	4.3108
1.24	0.81830	1.6272	1.1531	1.4112	0.98835	2.5263	1.73	0.63296	3.3250	1.4800	2.2467	0.84312	4.3547
1.25	0.81264	1.6562	1.1594	1.4286	0.98706	2.5568	1.74	0.63051	3.3655	1.4873	2.2629	0.83886	4.3989
1.26	0.80709	1.6855	1.1657	1.4460	0.98568	2.5876	1.75	0.62809	3.4062	1.4946	2.2791	0.83456	4.4433
1.27	0.80165	1.7150	1.1720	1.4634	0.98422	2.6187	1.76	0.62570	3.4472	1.5019	2.2952	0.83024	4.4880
1.28	0.79631	1.7448	1.1782	1.4808	0.98268	2.6500	1.77	0.62335	3.4884	1.5093	2.3113	0.82589	4.5330
1.29	0.79108	1.7748	1.1846	1.4983	0.98106	2.6816	1.78	0.62104	3.5298	1.5167	2.3273	0.82152	4.5783
1.30	0.78596	1.8050	1.1909	1.5157	0.97935	2.7135	1.79	0.61875	3.5714	1.5241	2.3433	0.81711	4.6238
1.31	0.78093	1.8354	1.1972	1.5331	0.97758	2.7457	1.80	0.61650	3.6133	1.5316	2.3592	0.81268	4.6695
1.32	0.77600	1.8661	1.2035	1.5505	0.97574	2.7783	1.81	0.61428	3.6554	1.5391	2.3751	0.80823	4.7155
1.33	0.77116	1.8970	1.2099	1.5680	0.97382	2.8112	1.82	0.61209	3.6978	1.5466	2.3909	0.80376	4.7618
1.34	0.76641	1.9282	1.2162	1.5854	0.97181	2.8444	1.83	0.60993	3.7404	1.5542	2.4067	0.79926	4.8083
1.35	0.76175	1.9596	1.2226	1.6028	0.96972	2.8778	1.84	0.60780	3.7832	1.5617	2.4224	0.79474	4.8551
1.36	0.75718	1.9912	1.2290	1.6202	0.96756	2.9115	1.85	0.60570	3.8262	1.5694	2.4381	0.79021	4.9022
1.37	0.75269	2.0230	1.2354	1.6376	0.96534	2.9455	1.86	0.60363	3.8695	1.5770	2.4537	0.78567	4.9498
1.38	0.74828	2.0551	1.2418	1.6550	0.96304	2.9798	1.87	0.60159	3.9130	1.5847	2.4693	0.78112	4.9974
1.39	0.74396	2.0874	1.2482	1.6723	0.96065	3.0144	1.88	0.59957	3.9568	1.5924	2.4848	0.77656	5.0453
1.40	0.73971	2.1200	1.2547	1.6896	0.95819	3.0493	1.89	0.59758	4.0008	1.6001	2.5003	0.77197	5.0934
1.41	0.73554	2.1528	1.2612	1.7070	0.95566	3.0844	1.90	0.59562	4.0450	1.6079	2.5157	0.76735	5.1417
1.42	0.73144	2.1858	1.2676	1.7243	0.95306	3.1198	1.91	0.59368	4.0894	1.6157	2.5310	0.76273	5.1904
1.43	0.72741	2.2190	1.2742	1.7416	0.95039	3.1555	1.92	0.59177	4.1341	1.6236	2.5463	0.75812	5.2394
1.44	0.72345	2.2525	1.2807	1.7589	0.94765	3.1915	1.93	0.58988	4.1790	1.6314	2.5615	0.75347	5.2886
1.45	0.71956	2.2862	1.2872	1.7761	0.94483	3.2278	1.94	0.58802	4.2242	1.6394	2.5767	0.74883	5.3381
1.46	0.71574	2.3202	1.2938	1.7934	0.94196	3.2643	1.95	0.58618	4.2696	1.6473	2.5919	0.74418	5.3878
1.47	0.71198	2.3544	1.3004	1.8106	0.93901	3.3011	1.96	0.58437	4.3152	1.6553	2.6070	0.73954	5.4378
1.48	0.70829	2.3888	1.3070	1.8278	0.93600	3.3382	1.97	0.58258	4.3610	1.6633	2.6220	0.73487	5.4880

TABLE E.4 (continued)

Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.98	0.58081	4.4071	1.6713	2.6369	0.73021	5.5385	2.55	0.50831	7.4196	2.1875	3.3918	0.47927	8.8505
1.99	0.57907	4.4534	1.6794	2.6518	0.72554	5.5894	2.56	0.50740	7.4792	2.1976	3.4034	0.47540	8.9162
2.00	0.57735	4.5000	1.6875	2.6666	0.72088	5.6405	2.57	0.50651	7.5391	2.2077	3.4149	0.47155	8.9821
2.01	0.57565	4.5468	1.6956	2.6814	0.71619	5.6918	2.58	0.50562	7.5992	2.2179	3.4263	0.46772	9.0482
2.02	0.57397	4.5938	1.7038	2.6962	0.71152	5.7434	2.59	0.50474	7.6595	2.2281	3.4376	0.46391	9.1146
2.03	0.57231	4.6411	1.7120	2.7109	0.70686	5.7952	2.60	0.50387	7.7200	2.2383	3.4489	0.46012	9.1813
2.04	0.57068	4.6886	1.7203	2.7255	0.70218	5.8473	2.61	0.50301	7.7808	2.2486	3.4602	0.45636	9.2481
2.05	0.56907	4.7363	1.7286	2.7400	0.69752	5.8997	2.62	0.50216	7.8418	2.2589	3.4714	0.45262	9.3154
2.06	0.56747	4.7842	1.7369	2.7545	0.69284	5.9523	2.63	0.50132	7.9030	2.2693	3.4825	0.44891	9.3829
2.07	0.56589	4.8324	1.7452	2.7690	0.68817	6.0052	2.64	0.50048	7.9645	2.2797	3.4936	0.44522	9.4507
2.08	0.56433	4.8808	1.7536	2.7834	0.68351	6.0584	2.65	0.49965	8.0262	2.2901	3.5047	0.44155	9.5187
2.09	0.56280	4.9295	1.7620	2.7977	0.67886	6.1118	2.66	0.49883	8.0882	2.3006	3.5157	0.43791	9.5869
2.10	0.56128	4.9784	1.7704	2.8119	0.67422	6.1655	2.67	0.49802	8.1504	2.3111	3.5266	0.43429	9.6553
2.11	0.55987	5.0275	1.7789	2.8216	0.66957	6.2194	2.68	0.49722	8.2128	2.3217	3.5374	0.43070	9.7241
2.12	0.55830	5.0768	1.7874	2.8402	0.66492	6.2736	2.69	0.49642	8.2754	2.3323	3.5482	0.42713	9.7932
2.13	0.55683	5.1264	1.7960	2.8543	0.66029	6.3280	2.70	0.49563	8.3383	2.3429	3.5590	0.42359	9.8625
2.14	0.55538	5.1762	1.8046	2.8683	0.65567	6.3827	2.71	0.49485	8.4014	2.3536	3.5697	0.42007	9.9320
2.15	0.55395	5.2262	1.8132	2.8823	0.65105	6.4377	2.72	0.49408	8.4648	2.3643	3.5803	0.41657	10.002
2.16	0.55254	5.2765	1.8219	2.8962	0.64644	6.4929	2.73	0.49332	8.5284	2.3750	3.5909	0.41310	10.072
2.17	0.55114	5.3270	1.8306	2.9100	0.64185	6.5484	2.74	0.49256	8.5922	2.3858	3.6014	0.40965	10.142
2.18	0.54976	5.3778	1.8393	2.9238	0.63728	6.6042	2.75	0.49181	8.6562	2.3966	3.6119	0.40622	10.212
2.19	0.54841	5.4288	1.8481	2.9376	0.63270	6.6602	2.76	0.49107	8.7205	2.4074	3.6224	0.40282	10.283
2.20	0.54706	5.4800	1.8569	2.9512	0.62812	6.7163	2.77	0.49033	8.7850	2.4183	3.6328	0.39945	10.354
2.21	0.54572	5.5314	1.8657	2.9648	0.62358	6.7730	2.78	0.48960	8.8497	2.4292	3.6431	0.39610	10.426
2.22	0.54440	5.5831	1.8746	2.9783	0.61905	6.8299	2.79	0.48888	8.9147	2.4402	3.6533	0.39276	10.498
2.23	0.54310	5.6350	1.8835	2.9918	0.61453	6.8869	2.80	0.48817	8.9800	2.4512	3.6635	0.38946	10,569
2.24	0.54182	5.6872	1.8924	3.0052	0.61002	6.9442	2.81	0.48746	9.0454	2.4622	3.6737	0.38618	10.641
2.25	0.54055	5.7396	1.9014	3.0186	0.60554	7.0018	2.82	0.48676	9.1111	2.4733	3.6838	0.38293	10.714
2.26	0.53929	5.7922	1.9104	3.0319	0.60106	7.0597	2.83	0.48607	9.1770	2.4844	3.6939	0.37970	10.787
2.27	0.53805	5.8451	1.9194	3.0452	0.59659	7.1178	2.84	0.48538	9.2432	2.4955	3.7039	0.37649	10.860
2.28	0.53683	5.8982	1.9285	3.0584	0.59214	7.1762	2.85	0.48470	9.3096	2.5067	3.7139	0.37330	10.933
2.29	0.53561	5.9515	1.9376	3.0715	0.58772	7.2348	2.86	0.48402	9.3762	2.5179	3.7238	0.37013	11.006
2.30	0.53441	6.0050	1.9468	3.0846	0.58331	7.2937	2.87	0.48334	9.4431	2.5292	3.7336	0.36700	11.080
2.31	0.53322	6.0588	1.9560	3.0976	0.57891	7.3529	2.88	0.48268	9.5102	2.5405	3.7434	0.36389	11.154
2.32	0.53205	6.1128	1.9652	3.1105	0.57452	7.4123	2.89	0.48203	9.5775	2.5518	3.7532	0.36080	11.228
2.33	0.53089	6.1670	1.9745	3.1234	0.57015	7.4720	2.90	0.48138	9.6450	2.5632	3.7629	0.35773	11.302
2.34	0.52974	6.2215	1.9838	3.1362	0.56580	7.5319	2.91	0.48074	9.7127	2.5746	3.7725	0.35469	11.377
2.35	0.52861	6.2762	1.9931	3.1490	0.56148	7.5920	2.92	0.48010	9.7808	2.5860	3.7821	0.35167	11.452
2.36	0.52749	6.3312	2.0025	3.1617	0.55717	7.6524	2.93	0.47946	9.8491	2.5975	3.7917	0.34867	11.527
2.37	0.52638	6.3864	2.0119	3.1743	0.55288	7.7131	2.94	0.47883	9.9176	2.6090	3.8012	0.34570	11.603
2.38	0.52528	6.4418	2.0213	3.1869	0.54862	7.7741	2.95	0.47821	9.986	2.6206	3.8106	0.34275	11.679
2.39	0.52419	6.4974	2.0308	3.1994	0.54438	7.8354	2.96	0.47760	10.055	2.6322	3.8200	0.33982	11.755
2.40	0.52312	6.5533	2.0403	3.2119	0.54015	7.8969	2.97	0.47699	10.124	2.6438	3.8294	0.33692	11.831
2.41	0.52206	6.6094	2.0499	3.2243	0.53594	7.9587	2.98	0.47638	10.194	2.6555	3.8387	0.33404	11.907
2.42	0.52100	6.6658	2.0595	3.2366	0.53175	8.0207	2.99	0.47578	10.263	2.6672	3.8479	0.33118	11.984
2.43	0.51996	6.7224	2.0691	3.2489	0.52758	8.0830	3.00	0.47519	10.333	2.6790	3.8571	0.32834	12.061
2.44	0.51894	6.7792	2.0788	3.2611	0.52344	8.1455	3.50	0.45115	14.125	3.3150	4.2608	0.21295	16.242
2.45	0.51792	6.8362	2.0885	3.2733	0.51932	8.2083	4.00	0.43496	18.500	4.0469	4.5714	0.13876	21.068
2.46	0.51691	6.8935	2.0982	3.2854	0.51521	8.2714	4.50	0.42355	23.458	4.8751	4.8119	0.09170	26.539
2.47	0.51592	6.9510	2.1080	3.2975	0.51112	8.3347	5.00	0.41523	29.000	5.8000	5.0000	0.06172	32.654
2.48	0.51493	7.0088	2.1178	3.3095	0.50706	8.3983	6.00	0.40416	41.833	7.941	5.2683	0.02965	46.815
2.49	0.51395	7.0668	2.1276	3.3214	0.50303	8.4622	7.00	0.39736	57.000	10.469	5.4444	0.01535	63.552
2.50	0.51299	7.1250	2.1375	3.3333	0.49902	8.5262	8.00	0.39289	74.500	13.387	5.5652	0.00849	82.865
2.51	0.51204	7.1834	2.1474	3.3451	0.49502	8.5904	9.00	0.38980	94.333	16.693	5.6512	0.00496	104.753
2.52	0.51109	7.2421	2.1574	3.3569	0.49104	8.6549	10.00	0.38757	116.50	20.388	5.7143	0.00304	129.217
2.53	0.51015	7.3010	2.1674	3.3686	0.48709	8.7198	∞	0.37796	∞	∞	6.000	0	∞
2.54	0.50923	7.3602	2.1774	3.3802	0.48317	8.7850							